Notes on Computational Fluid Dynamics: General Principles

5.12 Steady-state solution

The equations in Sec. 5.10 are combined using algorithms that couple the solutions for p and $\mathbf{u}$. One algorithm is SIMPLE (Semi-Implicit Method for Pressure-Linked Equations)[3], which is presented here with a modern interpretation.

The SIMPLE algorithm is generally used to produce steady flow solutions in CFD. These solutions are directly applicable for flows that reach a *steady state*, *i.e.* when flow variables stop changing in time. They can also provide approximate solutions to flows that are moderately unsteady, usually at a lower cost than a more exact transient solution.

An example of the algorithm is shown for the system of equations presented in Sec. 5.9. The time derivative $(\partial/\partial t)$ terms are omitted due to the steady-state assumption.

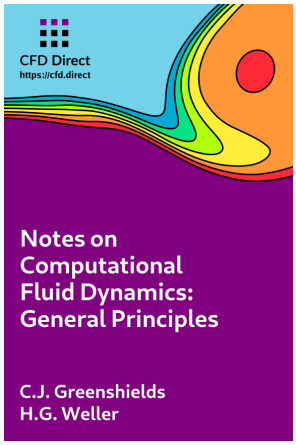
The algorithm involves an iterative sequence with steps, $i = 1, 2, \ldots, i_{\text{end}}$. It begins by constructing a matrix equation for energy which is under-relaxed by a factor α_T. The equation is solved for T, which is used to update $\rho(T, p)$ according to an equation of state. A matrix equation is then constructed using all the terms from the momentum equation *excluding* ∇p, *i.e.*

$$\mathbf{A} \bullet \mathbf{u} - \mathbf{b} \equiv \nabla \bullet (\mathbf{uu}) - \nabla \bullet (\nu \nabla \mathbf{u}) - \frac{\rho(T,p)}{\rho_0}\mathbf{g} \tag{5.21}$$

The matrix equation is under-relaxed by a factor $\alpha_{\mathbf{u}}$ before equating with $-\nabla p$ and solving for $\mathbf{u}$ (the *momentum predictor*).

$A\mathbf{u}$ and $\mathbf{H}(\mathbf{u})$ are then evaluated from $\mathbf{A} \bullet \mathbf{u} - \mathbf{b}$ (the *momentum matrix*), as described on page 351. They are used to form the *pressure equation*, which is solved for p.

The new pressure p is used to correct the flux ϕ_{f} so that it obeys mass conservation better (the *flux corrector*). It is then under-relaxed by a

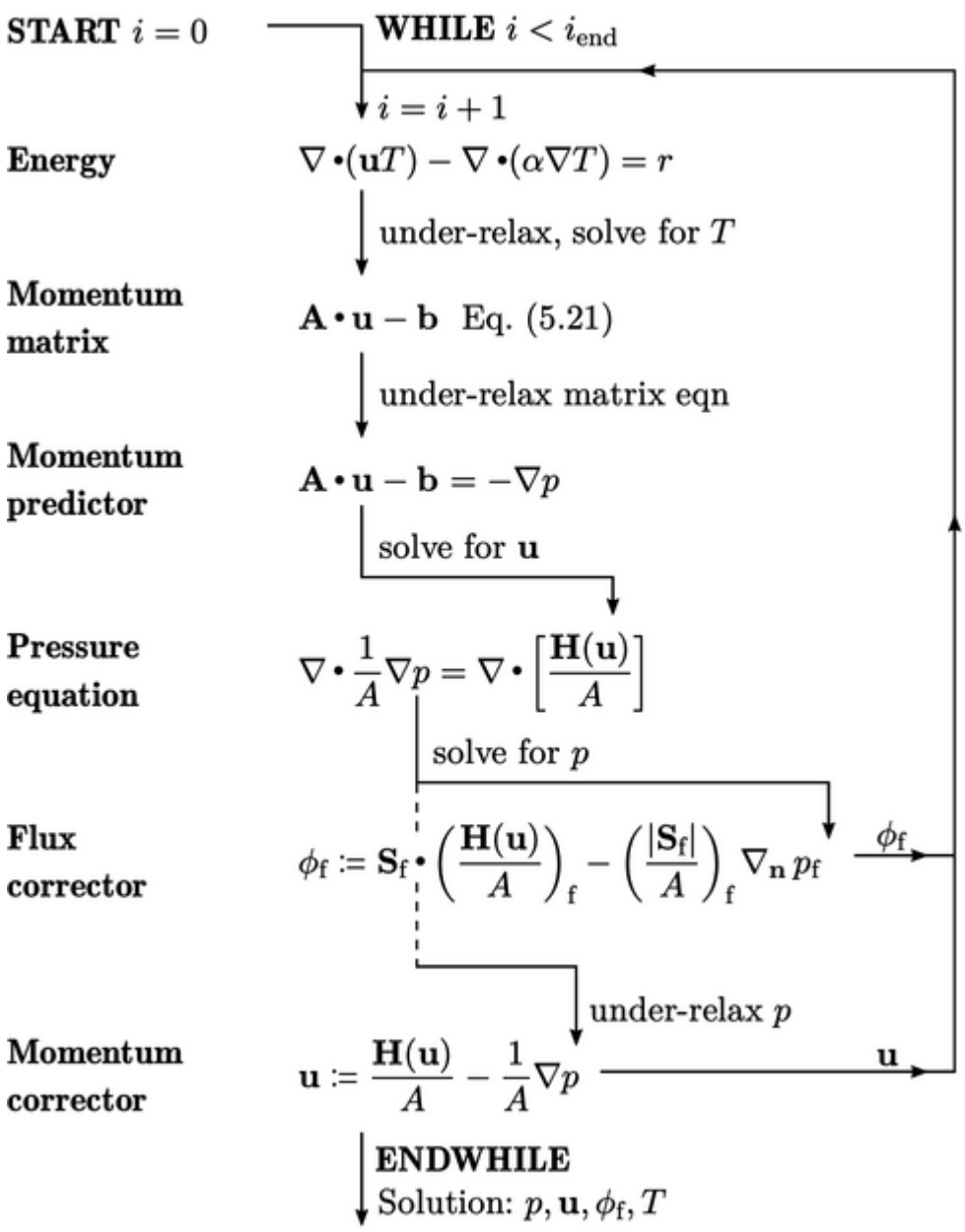


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factor α_p before correcting $\mathbf{u}$ before the next solution step begins (the *momentum corrector*).



[3] Suhas Patankar and Brian Spalding, *A calculation procedure for heat, mass and momentum transfer in three-dimensional parabolic flows*, 1972.

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